LECTURE SHEET

EXERCISE-I

Continuity

LEVEL-I (MAIN)

Single answer type questions

1. Let k be a non-zero real number. If $f(x)=\begin{cases}\dfrac{(e^x-1)}{\sin\left(\dfrac{x}{k}\right)\log\left(1+\dfrac{x}{4}\right)}, & x\neq 0\\ 12 & ,x=0\end{cases}$ is a continuous function then the vlaue of k is:

   1) 4 2) 1 3) 3 4) 2

2. If $f(x)=\dfrac{3x+\tan^2 x}{x}$ is continuous at $x = 0$, then $f(0)$ is equal to

   1) 3 2) 2 3) 4 4) 0

3. If $f(x)=\dfrac{\log(1+ax)-\log(1-bx)}{x}$ for $x\neq 0$ and $f(0) = k$ and $f(x)$ is continuous at $x = 0$, then k is equal to

   1) $a + b$ 2) $a - b$ 3) a 4) b

4. The function $f(x)=|x|$ at $x = 0$ is

   1) continuous but non-differentiable 2) discontinuous and differentiable
   3) discontinuous and non-differentiable 4) continuous and differentiable

5. In order that the function $f(x)=(x+1)^{1/x}$ is continuous at $x = 0$, $f(0)$ must be defined as

   1) $f(0) = 0$ 2) $f(0) = e$ 3) $f(0) = 1/e$ 4) $f(0) = 1$

6. If $f(x)=\begin{cases}\left[\tan\left(\dfrac{\pi}{4}+x\right)\right]^{1/x} & x\neq 0\\ k, & x=0\end{cases}$. For what values of k, $f(x)$ is continuous at $x = 0$?

   1) 1 2) 0 3) e 4) e^2

7. The function $f(x)=\dfrac{1}{x^2-3|x|+2}$ is discontinuous at the points

   1) $x = 1, 2$ 2) $x=\pm 1,\pm 2$ 3) R 4) $R - \{1,2\}$

9. Let a function f be defined by $f(x)=\frac{x-|x|}{x}$ for $x\neq 0$ and $f(0)=2$, then f is

1) continuous no where
2) continuous every where
3) continuous for all x except $x = 1$
4) continuous for all x except $x = 0$

10. The function $f:R-\{0\}\to R$ given by $f(x)=\frac{1}{x}-\frac{2}{e^{2x}-1}$ can be made continuous at $x = 0$ by defining $f(0)$ as

1) 2
2) –1
3) 0
4) 1

11. The value of p and q for which the function $f(x)=\begin{cases}\frac{\sin(p+1)x+\sin x}{x} & x<0\\ q, & x=0\\ \frac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}} & x>0\end{cases}$ is continuous for all x are

1) $p=\frac{1}{2},q=\frac{3}{2}$
2) $p=\frac{1}{2},q=-\frac{3}{2}$
3) $p=\frac{5}{2},q=\frac{1}{2}$
4) $p=-\frac{3}{2},q=\frac{1}{2}$

12. Let $f:(1,\infty)\to(0,\infty)$ be a continuous decreasing function with $\lim_{x\to\infty}\frac{f(4x)}{f(8x)}=1$. Then $\lim_{x\to\infty}\frac{f(6x)}{f(8x)}$ is equal to

1) $\frac{4}{8}$
2) $\frac{4}{6}$
3) $\frac{6}{8}$
4) 1

13. If $f(x)=\begin{cases}\frac{1-\sin^3 x}{3\cos^2 x}, & if\ x<\frac{\pi}{2}\\ a, & if\ x=\frac{\pi}{2}\\ \frac{b(1-\sin x)}{(\pi-2x)^2}, & if\ x>\frac{\pi}{2}\end{cases}$, so that $f(x)$ is continuous at $x=\frac{\pi}{2}$, then

1) $a = 1/2$
2) $b = 5$
3) $a = 1$
4) $b = -4$

14. If $f(x)=\frac{\tan\left(\frac{\pi}{4}-x\right)}{\cot 2x},(x\neq\pi/4)$, is continuous at $x=\pi/4$, then the value of $f\left(\frac{\pi}{4}\right)$ is

1) 1
2) 1/2
3) 1/3
4) –1

15. The function f is defined in $[-5,5]$ as $f(x) = x$, if x is rational and $f(x) = -x$, if x is irrational. Then

1) $f(x)$ is continuous at every x, except $x = 0$.
2) $f(x)$ is discontinuous at every x, except $x = 0$.
3) $f(x)$ is continuous everywhere.
4) $f(x)$ is discontinuous everywhere.

16. If $f(x)=\begin{cases}\dfrac{|x+2|}{\tan^{-1}(x+2)}, & if\ x\neq -2\\ 2\ , & if\ x=-2\end{cases}$ then, $f(x)$ is

1) continuous at $x = -2$
2) not continuous at $x = -2$
3) differentiable at $x = -2$
4) continuous but not diferentiable at $x = -2$

17. Let $f(x)=\dfrac{1-\tan x}{4x-\pi}, x\neq\dfrac{\pi}{4}, x\in\left[0,\dfrac{\pi}{2}\right]$. If $f(x)$ is continuous in $[0,\pi/2]$, then $f(\pi/4)$ is

1) 1
2) $\dfrac{1}{2}$
3) $\dfrac{-1}{2}$
4) -1

18. If $f(x)=\begin{cases}x^{a}\sin\left(\dfrac{1}{x}\right), & x\neq 0\\ 0, & x=0\end{cases}$ is continuous but non-differentiable at $x = 0$, then

1) $a\in(-1,0)$
2) $a\in(0,2]$
3) $a\in(0,1]$
4) $a\in[1,2)$

19. If $f:R\to R$ is a function defined by $f(x)=[x]\cos\left(\dfrac{2x-1}{2}\right)\pi$, where $[x]$ denotes the greatest integer function, then f is

1) continuous for every real x.
2) discontinuous only at $x = 0$.
3) discontinuous only at non-zero integral values of x.
4) continuous only at $x = 0$.

20. Let $f(x)$ be defined in the interval $[0,4]$ such that $f(x)=\begin{cases}1-x, & 0\leq x\leq 1\\ x+2, & 1<x<2\\ 4-x, & 2\leq x\leq 4\end{cases}$ Then the number of points where $f(f(x))$ is discontinuous is

1) 1
2) 2
3) 3
4) 4

21. If $f(x)=\begin{cases}x+2, & x<0\\ -x^2-2, & 0\leq x<1\\ x & x\geq 1\end{cases}$, then the number of points of discontinuity of $|f(x)|$ is

1) 1
2) 2
3) 3
4) 4

22. The set of points of discontinuity of the function $f(x)=\lim\limits_{n\to\infty}\dfrac{(2\sin x)^{2n}}{3^n-(2\cos x)^{2n}}$ is given by

1) R
2) $\left\{n\pi\pm\dfrac{\pi}{3}, n\in I\right\}$
3) $\left\{n\pi\pm\dfrac{\pi}{6}, n\in I\right\}$
4) $n\pi, n\in I$

23. Let $f(x)=[2x^3-6]$ when $[x]$ is greatest integer less than or equal to x then the number of points in ... tinuous is

3) 13
4) 12

24. If [.] denotes the greatest integer function then the number of points where $f(x)=[x]+\left[x+\frac{1}{3}\right]+\left[x+\frac{2}{3}\right]$ is discontinuous for $x\in(0,3)$ are

1) 2 2) 9 3) 8 4) 10

25. The function $f(x)=\{x\}\sin(\pi[x])$, where [.] denotes the greatest integer function and {.} is the fractional part function, is discontinuous at

1) all x
2) all integer points
3) no x
4) x which is not an integer

26. The function $f(x)=|2\,\mathrm{sgn}(2x)|+2$ has

1) Jump discontinuity
2) Removable discontinuity
3) Infinite discontinuity
4) No discontinuity at $x=0$

27. If $f(x)=Sgn(2\sin x+a)$ is continuous for all x then the possible values of 'a' are

1) R 2) $a<-2$ or $a>2$ 3) $(-2,2)$ 4) $(0,\infty)$

LEVEL-II (ADVANCED)

Single answer type questions

1. If $f(x)=\begin{cases} xe^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}, & x\neq 0 \\ 0, & x=0 \end{cases}$ then $f(x)$ is

a) continuous as well as differentiable for all x.

b) continuous for all x but not differentiable at $x = 0$

c) neither differentiable nor continuous at $x = 0$.

d) discontinuous everywhere.

2. Let $f(x)=\begin{cases} \frac{x-4}{|x-4|}+a, & x<4 \\ a+b, & x=4 \\ \frac{x-4}{|x-4|}+b, & x>4 \end{cases}$ Then $f(x)$ is continuous at $x = 4$ when

a) $a = 0, b = 0$ b) $a = 1, b = 1$ c) $a = -1, b = 1$ d) $a = 1, b = -1$

3. Let [.]denotes the greatest integer function and $f(x)=\left[\tan^2 x\right]$. Then

a) $\lim_{x\to 0} f(x)$ does not exist
b) $f(x)$ is continuous at $x = 0$
c) $f(x)$ is not differentiable at $x = 0$
d) $f'(0)=1$

4. Let f and g be real valued functions defined on interval $(-1,1)$ such that $g''(x)$ is continuous, $g(0) \neq 0, g'(0) = 0, g''(0) \neq 0$, and $f(x) = g(x)\sin x$.

Statement 1 : $\lim\limits_{x \to 0}[g(x)\cot x - g(0)\operatorname{cosec} x] = f''(0)$

Statement 2 : $f'(0) = g(0)$

a) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.

b) Statement 1 is true, Statement 2 is true; Statement 2 is NOT a correct explanation for Statement 1

c) Statement 1 is true, statement 2 is false.

d) Statement 1 is false, Statement 2 is true.

5. The value of k for which $f(x) = \begin{vmatrix} \left(1 + xe^{-1/x^2}\sin\dfrac{1}{x^4}\right)^{e^{1/x^2}}, & x \neq 0 \\ k, & x = 0 \end{vmatrix}$ is continuous at $x = 0$, is

a) 4 b) 3 c) 2 d) 1

6. Consider [.] and {.} denote the greatest integer function and the fractional part function, respectively. Let $f(x) = \{x\} + \sqrt{\{x\}}$

Statement I - f is not differentiable at integral values of x.

Statement II - f is not continuous at integral points.

a) Both I and II are individually true and II is the correct expalnation of I

b) Both I and II are individually true but II is not the correct explanation of I

c) I is true but II is false

d) I is false but II is true

7. Statement I - Consider the functions $f(x) = x^2 + |2x|$.

Statement II - $F'(0^+) = 2$ and $F'(0^-) = -2$

a) Both I and II are individually true and II is the correct expalnation of I

b) Both I and II are individually true but II is not the correct explanation of I

c) I is true but II is false

d) I is false but II is true

9. Let $[x]$ denotes the greatest integer less than or equal to x. If $f(x)=[x\sin\pi x]$, then $f(x)$ is

a) continuous at $x = 0$
b) contiuous in $(-1, 0)$
c) differentiable at $x = 1$
d) differentiable in $(-1,1)$

10. The function $f(x)=\begin{cases} |x-3|, & x\geq 1 \\ \frac{x^2}{4}-\frac{3x}{2}+\frac{13}{4}, & x<1 \end{cases}$ is

a) continuous at $x = 1$
b) differentiable at $x = 1$
c) continuous at $x = 3$
d) differentiable at $x = 3$

11. Which of the following funcitons are continus on $(0,\pi)$?

a) $\tan x$
b) $\int_0^x t\sin\frac{1}{t}dt$
c) $\begin{cases} 1, & 0<x\leq\frac{3\pi}{4} \\ 2\sin\frac{2}{9}x, & \frac{3\pi}{4}<x<\pi \end{cases}$
d) $\begin{cases} x\sin x, & 0<x\leq\frac{\pi}{2} \\ \frac{\pi}{2}\sin(\pi+x), & \frac{\pi}{2}<x<\pi \end{cases}$

12. The function $f(x) = \left||e^x-1|-1\right|$ is

a) continuous for all x
b) differentiable for all x
c) not continuous at $x = 0$, ln 2
d) not differentiable at $x = ln$ 2

13. Let $f(x)=\begin{cases} 0, & x<0 \\ x^2, & x\geq 0 \end{cases}$. Then for all x,

a) f' is differentiable
b) f is differentiable
c) f' is continuous
d) f is continuous

14. Let $f(x)=(x+|x|)|x|$, then for all x

a) f is continuous
b) f' is differentiable, $\forall x\in R$
c) f' is continuous
d) f " is continuous

15. Let $h(x)=\min\{x,x^2\}$, for every real number of x. Then

a) h is continuous for all x
b) h is differentiable for all x
c) $h'(x) = 1$ for all $x > 1$
d) h is not differentiable at two valuels of x

16. Which of the following function(s) has/have removable discontinuity at $x = 1$?

a) $f(x)=\frac{1}{\ln|x|}$
b) $f(x)=\frac{x^2-1}{x^3-1}$
c) $f(x)=2^{-2^{\frac{1}{1-x}}}$
d) $f(x)=\frac{\sqrt{x+1}-\sqrt{2x}}{x^2-x}$

17. If $f(x) = \min\{\tan x, \cot x\}$, then

a) $f(x)$ is discontinuous at $x = 0, \frac{\pi}{4}, \frac{5\pi}{4}$

b) $f(x)$ is continuous at $x = 0, \frac{\pi}{2}, \frac{3\pi}{2}$

c) Range of $f(x)$ is $(-\infty,-1]\cup[0,1]$

d) $f(x)$ is periodic with period π

18. $f(x)=\begin{cases} \left(\frac{3}{2}\right)^{(\cot 3x)/(\cot 2x)}, & 0<x<\frac{\pi}{2} \\ b+3, & x=\frac{\pi}{2} \\ (1+|\cot x|)^{(a|\tan x|)/b}, & 0<x<\frac{\pi}{2} \end{cases}$ is continuous at $x=\pi/2$. Then

a) $a = 0$ b) $a = 2$ c) $b = -2$ d) $b = 2$

19. Let $f(x)=\begin{cases} \frac{e^x-1+ax}{x^2}, & x>0 \\ b, & x=0 \\ \frac{\sin\frac{x}{2}}{x}, & x<0 \end{cases}$, Then

a) $f(x)$ is continuous at $x = 0$ if $a=-1, b=\frac{1}{2}$

b) $f(x)$ is discontinuous at $x = 0$ if $b\neq\frac{1}{2}$

c) $f(x)$ has irremovable discontinuity at $x = 0$ if $a\neq-1$

d) $f(x)$ has removable discontinuity at $x = 0$ if $a = -1$, $a\neq\frac{1}{2}$

20. If $f(x) = \min\{1, x^2, x^3\}$, then

a) $f(x)$ is continuous everywhere

b) $f(x)$ is continuous and differentiable everywhere

c) $f(x)$ is not differentiable at two points

d) $f(x)$ is not differentiable at one point

21. The function, $f(x) = \max\{(1-x), (1+x), 2\}$, $x\in(-\infty, \infty)$ is

a) continuous at all points

b) differentiable at all points

c) differentiable at all points except at $x = 1$ and $x = -1$

d) continuous at all points except at $x = 1$ and $x = -1$, where it is discontinuous

22. Let $L=\lim_{x\to 0}\frac{a-\sqrt{a^2-x^2}-\frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then

a) $a = 2$ b) $a = 1$ c) L = 1/64 d) L = 1/34.

23. Let $f: R \to R$ be a functin such that $f(x+y) = f(x) + f(y), \forall\, x, y \in R$. If $f(x)$ is differentialbe at $x = 0$, then

a) $f(x)$ is differentiable only in a finite interval containing zero

b) $f(x)$ is continuous $\forall\, x \in R$

c) $f'(x)$ is continuous $\forall\, x \in R$

d) $f(x)$ is differentiable except at finitely many points

24. If $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \le -\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \le 0 \\ x - 1, & 0 < x \le 1 \\ \ln x, & x > 1 \end{cases}$, then

a) $f(x)$ is continuous at $x = -\frac{\pi}{2}$

b) $f(x)$ is not differentialbe at $x = 0$

c) $f(x)$ is differentialbe at $x = 1$

d) $f(x)$ is differentialbe at $x = -3/2$

25. For every integer n, let a_n and b_n be real numbers. Let function $f: R \to R$ be given by
$f(x) = \begin{cases} a_n + \sin \pi x, & for\ x \in [2n, 2n+1] \\ b_n + \cos \pi x, & for\ x \in (2n-1, 2n) \end{cases}$, for all integers n. If f is continuous, then which of the following hold(s) for all n

a) $a_{n-1} - b_{n-1} = 0$

b) $a_n - b_n = 1$

c) $a_n - b_{n+1} = 1$

d) $a_{n-1} - b_n = -1$

Linked comprehension type questions

Passage - I :

Let $f(x) = \begin{cases} \frac{a(1 - x\sin x) + b\cos x + 5}{x^2}, & x < 0 \\ 3 & x = 0 \end{cases}$, *Where* $P(x)$ *is a cubic*

EXERCISE-II

Differentiability

LEVEL-I (MAIN)

Single answer type questions

1. $\sin^{-1}\left(\frac{1+x^2}{2x}\right)$ is

1) Continuous but not differentiable at $x = 1$
2) Differentiable at $x = 1$
3) Neither continuous nor differentiable at $x = 1$
4) Continuous every where

2. For $x \in R, f(x) = |\log_e 2 - \sin x|$ and $g(x) = f(f(x))$, then

1) $g'(0) = \cos(\log_e 2)$
2) $g'(0) = -\cos(\log_e 2)$
3) g is differentiable at $x = 0$ and $g'(0) = -\sin(\log_e 2)$
4) g is not differentiable at $x = 0$

3. $f(x) = \begin{cases} x^2\left(\dfrac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then

1) $f(x)$ is discontinuous at $x = 0$
2) $f(x)$ is continuous but non-differentiable at $x = 0$
3) $f(x)$ is differntiable at $x = 0$
4) $f'(0) = 2$

4. Suppose $f(x)$ is differentiable at $x = 1$ and $\lim\limits_{h \to 0} \frac{1}{h} f(1+h) = 5$, then $f'(1)$ equals

1) 3 2) 4 3) 5 4) 6

5. If f is a real-valued differentiable function satisfying $|f(x) - f(y)| \leq (x-y)^2, x, y \in R$ and $f(0) = 0$, then $f(1)$ equals

1) -1 2) 0 3) 2 4) 1

6. The set of points where $f(x) = \frac{x}{1+|x|}$ is differentiable is

1) $(-\infty, 0) \cup (0, \infty)$ 2) $(-\infty, -1) \cup (-1, \infty)$ 3) $(-\infty, \infty)$ 4) $(0, \infty)$

7. If $f(x) = \begin{cases} x^3, & x^2 < 1 \\ x & x^2 \geq 1 \end{cases}$, then $f(x)$ is differentiable at

1) $(-\infty, \infty) - \{1\}$ 2) $(-\infty, \infty) - \{-1, 1\}$ 3) $(-\infty, \infty) - \{1 - 1, 0\}$ 4) $(-\infty, \infty) - \{-1\}$

8. Let $f: R \to R$ be a function defined by $f(x) = \min\{x+1, |x|+1\}$. Then which of the following is true?

1) $f(x) \geq 1$ for all $x \in R$.
2) $f(x)$ is not differentiable at $x = 1$.
3) $f(x)$ is differentiable everywhere.
4) $f(x)$ is not differentiable at $x = 0$.

9. The function $f(x) = |x^3|$ is

1) differentiable everywhere
2) continuous but not differentiable at x = 0
3) not a continuous function
4) no where differentiable

10. Let $f(x) = x|x|$ and $g(x) = \sin x$

Statement - 1: gof is differferntiable at $x = 0$ and its derivative is continuous at that point.

Statement - 2 : gof is twice differentiable at $x = 0$

1) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.
2) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1.
3) Statement 1 is true, statement 2 is false
4) Statement 1 is false, statement 2 is true

11. If the function $g(x) = \begin{cases} k\sqrt{x+1}, & 0 \leq x \leq 3 \\ mx+2, & 3 < x \leq 5 \end{cases}$ is differentiable, then the value of $k + m$ is

1) 2
2) $\frac{16}{5}$
3) $\frac{10}{3}$
4) 4

12. For a real number y, let $[y]$ denotes the greatest integer less than or equal to y. Let $f(x) = \frac{\tan(\pi[x-\pi])}{1+[x]^2}$ Then

1) $f(x)$ is discontinuous at some x
2) $f(x)$ is continuous at all x, but the derivative $f'(x)$ does not exist for some x.
3) $f'(x)$ exist for all x, but the derivative $f'(x_0)$ does not exist for some x
4) $f'(x)$ exists for all x

13. If $f(x) = \begin{cases} (\cos x)^{1/\sin x} & for\ x \neq 0 \\ k & for\ x = 0 \end{cases}$ the value of k, so that f is differentiable at $x = 0$ is

1) 0
2) 1
3) $1/2$
4) 2

14. Let $f(x) = \begin{cases} \frac{\sin|x^2-5x+6|}{x^2-5x+6}, & x \neq 2,3 \\ 1, & x = 2 or 3 \end{cases}$ the set of all points where f is differentiable is

1) $(-\infty, \infty)$
2) $(-\infty, \infty) - \{2\}$
3) $(-\infty, \infty) - \{3\}$
4) $(-\infty, \infty) - \{2, 3\}$

15. Let $f: R \to R$ be a function defined by $f(x) = \max\{x, x^3\}$. Then set of all points where $f(x)$ is not differentiable is

1) $\{-1, 1\}$
2) $\{-1, 0\}$
3) $\{0, 1\}$
4) $\{-1, 0, 1\}$

16. The function given by $y=||x|-1|$ is differentiable for all real numbers except the points

1) $\{0,1,-1\}$ 2) ± 1 3) 1 4) -1

17. If $f(x)$ is a continuous and differentiable function and $f(1/n)=0\ \forall n\geq 1$ and $n\in I$, then

1) $f(x)=0, x\in(0,1]$
2) $f(0)=0, f'(0)=0$
3) $f(0)=0=f'(0), x\in(0,1]$
4) $f(0)=0$ and $f'(0)$ need not be zero

18. Let $f(x)=\begin{cases} x^2\left|\cos\frac{\pi}{x}\right|, & x\neq 0 \\ 0, & x=0 \end{cases}, x\in I$. Then f is

1) differentiable both at $x=0$ and at $x=2$
2) differentiable at $x=0$ but not differentiable at $x=2$
3) not differentiable at $x=0$ but differentiable at $x=2$
4) differentiable neither at $x=0$ nor at $x=2$

19. Let $f:R\to R$ be a funtction defined by $f(x)=\min\{x+1,|x|+1\}$. Then which of the following is true?

1) $f(x)$ is differentiable everywhere
2) $f(x)$ is not diffferentiable at $x=0$
3) $f(x)\geq 1$ for all $x\in R$
4) $f(x)$ is not differentaible at $x=1$

20. $f(x)=\max\{x/n,|\sin\pi x|\}, n\in N$, has maximum points of non- differentiability for $x\in(0,4)$. Then n cannot be

1) 4 2) 2 3) 5 4) 6

21. A function $f(x)$ is defined as $f(x)=\begin{cases} x^m\sin\frac{1}{x}, & x\neq 0, m\in N \\ 0, & if\ x=0 \end{cases}$

The least value of m for which $f(x)$ is continuous at $x=0$ is

1) 1 2) 2 3) 3 4) none

LEVEL-II (ADVANCED)

Single answer type questions

1. Let $a,b\in R$ and $f:R\to R$ be defined by $f(x)=a\cos\left(|x^3-x|\right)+b|x|\left(\sin|x^3+x|\right)$. Then f is

a) differentiable at $x=0$ if $a=0$ and $b=1$
b) differentiable at $x=1$ if $a=1$ and $b=2$
c) NOT differentiable at $x=0$ if $a=1$ and $b=0$
d) NOT differentiable at $x=1$ if $a=1$ and $b=1$

2. If $f(x)=(x^2-4)\left|x^3-6x^2+11x-6\right|+\frac{x}{1+|x|}$, then the set of points at which the function $f(x)$ is not differentiable is

a) $\{-2,2,1,3\}$ b) $\{-2,0,3\}$ c) $\{-2,2,0\}$ d) $\{1,3\}$

3. Let $f(x)=(x-1)\sin\left(\frac{1}{x-1}\right)$. Then which one of the following is true?

a) f is neither differntiable at $x = 0$ nor at $x = 1$
b) f is differentiable at $x = 0$ and at $x = 1$.
c) f is differentiable at $x = 0$ but not at x=0, x=1
d) f is differentiable at $x = 1$ but not at $x = 0$

4. The function $f(x)=(x^2-1)\left|x^2-3x+2\right|+\cos(|x|)$ is not differentiable at

a) -1 b) 0 c) 1 d) 2

5. The left-hand derivative of $f(x)=[x]\sin(\pi x)$ at $x = k$, k is integer, is

a) $(-1)^k(k-1)\pi$ b) $(-1)^{k-1}(k-1)\pi$ c) $(-1)^k k\pi$ d) $(-1)^{k-1}k\pi$

6. Which of the following functions is differentiable at $x = 0$?

a) $\cos(|x|)+|x|$ b) $\cos(|x|)-|x|$ c) $\sin(|x|)+|x|$ d) $\sin(|x|)-|x|$

7. Let 'f' be a function defined by $2f(\sin x)+f(\cos x)=x\,\forall x$, then set of points where 'f' is not differentiable is

a) Set of all natural numbers
b) Set of all irrational numbers
c) $\{0,1,-1\}$
d) $\{1,-1\}$

8. The domain of derivative of the following function is $f(x)=\begin{cases}\tan^{-1}x, & \text{if } |x|\le 1\\ \frac{1}{2}(|x|-1) & \text{if } |x|>1\end{cases}$

a) $R-\{0\}$ b) $R-\{1\}$ c) $R-\{-1\}$ d) $R-\{-1,1\}$

9. Let $f(x) = x^3 - x^2 + x + 1$ and $g(x)$ be a function defined by $g(x)=\begin{cases}\max\{f(t):0\le t\le x\}, & 0\le x\le 1\\ 3-x, & 1<x\le 2\end{cases}$, then $g(x)$ is

a) continuous and differentiable on [0,2]
b) continuous but not differentiable on [0,2]
c) Neither continuous nor differentiable on [0,2]
d) none of the above

11. If $f(x) = x^3 \operatorname{sgn} x$, then

a) f is derivable at $x = 0$
b) f is continuous but not derivable at $x = 0$
c) LHD at $x = 0$ is 1
d) RHD at $x = 0$ is 1

More than one correct answer type questions

12. The function $f(x)=1+|\sin x|$ is

a) continuous no where
b) continuous everywhere
c) not differentiable at infinite number of point
d) not differentiable at $x = 0$

13. Which of the following is/are true for $f(x)=\operatorname{sgn}(x)\times\sin x$?

a) Discontinuous no where
b) An even funciton
c) $f(x)$ is periodic
d) $f(x)$ is differentiable for all x

14. If $f(x)=[|x|]$, where [.]denotes the greatest integer function, then which of the following is not true?

a) $f(x)$ is continuous $\forall x \in R$
b) $f(x)$ is continuous from right and discontinous from left $\forall x \in N$
c) $f(x)$ is continuous from left and discontinuous from right $\forall x \in I$
d) $f(x)$ is continous at $x = 0$

15. If $f(x)=\frac{x-2}{2}$, then in $[0,\pi]$,

a) both $\tan(f(x))$ and $\frac{1}{f(x)}$ are continuous
b) $\tan(f(x))$ is continuous but $f^{-1}(x)$ is not continuous
c) $\tan(f^{-1}(x))$ and $f^{-1}(x)$ are discontinuous
d) every where discontinuous

16. Let $g(x)=x\,f(x)$, where $f(x)=\begin{cases} x\sin\frac{1}{x}, & x\neq 0 \\ 0, & x=0 \end{cases}$. At $x=0$,

a) g is differentiable but g' is not continuous
b) g is differentiable while f is not
c) both f and g are differentiable
d) g is differentiable and g' is continuos

17. Let $f(x)=\begin{cases} \frac{\sin[x^2]\pi}{x^2-2x-18}+ax^3+b & \text{for } 0\le x\le 1 \\ 2\cos\pi x+\tan^{-1}x & \text{for } 1<x\le 2 \end{cases}$ is differentiable in [0, 2], $[x]$ denotes the greatest integer less than or equal to x, then

a) $a=\frac{1}{6}$
b) $a=\frac{1}{4}$
c) $b=\frac{\pi}{4}-\frac{11}{6}$
d) $b=\frac{\pi}{4}-\frac{13}{6}$

18. If the function $f(x)=\left[\frac{(x-2)^3}{a}\right]\sin(x-2)+a\cos(x-2)$ (where $[x]$ denotes the greatest ...

19. If $f(x)=\cos\left\{\frac{\pi}{2}[x]-x^3\right\}, -1<x<2$ and $[x]$ is the greatest integer less than or equal to x, then $f\left(\sqrt[3]{\frac{\pi}{2}}\right)$ is

a) a rational number b) an irrational number c) 0 d) $\sqrt{2}$

20. If $f(x)=x+|x|+\cos([\pi^2]x)$ and $g(x)=\sin x$, where[.] denotes the greatest integer function, then

a) $f(x)+g(x)$ is continuous everywhere

b) $f(x)+g(x)$ is differentiable everywhere

c) $f(x)\times g(x)$ is differentiable everywhere

d) $f(x)\times g(x)$ is continuous but not differentiable at $x=0$

21. Which of the following function is thrice differentiable at $x=0$?

a) $f(x)=|x^3|$ b) $f(x)=x^3|x|$ c) $f(x)=|x|\sin^3 x$ d) $f(x)=x|\tan^3 x|$

Linked comprehension type questions

Passage - I :

Let $f(x)=\begin{cases} x+2 & 0\le x<2 \\ 6-x, & x\ge 2\end{cases}$ $g(x)=\begin{cases} 1+\tan x, & 0\le x<\frac{\pi}{4} \\ 3-\cot x, & \frac{\pi}{4}\le x<\pi\end{cases}$

22. $f(g(x))$ is

a) discontinuous at $x=\pi/4$ b) differentiable at $x=\pi/4$

c) continuous but non-differentiable at $x=\pi/4$

d) differentiable at $x=\pi/4$, but derivative is not continuous

23. The number of points of non-differentiability of $h(x)=|f(g(x))|$ is

a) 1 b) 2 c) 3 d) 4

24. The range of $h(x)=f(g(x))$ is

a) $(-\infty,\infty)$ b) $(4,\infty)$ c) $(-\infty,4]$ d) none of these

Passage - III :

Given the continuous function $y=f(x)=\begin{cases} x^2+10x+8, & x\le -2 \\ ax^2+bx+c & -2<x<0, a\ne 0 \\ x^2+2x & x\ge 0\end{cases}$

If a line L touches the graph of $y=f(x)$ at three points , then

25. The slope of the line L is equal to

a) 1 b) 2 c) 4 d) 6

26. The value of $(a+b+c)$ is equal to

a) $5\sqrt{2}$ b) 5 c) 6 d) 7

27. If ... $f(x)$ is differentiable at $x=0$, then the value of b

Integer answer type questions

28. Let $f:R\to R$ and $g:R\to R$ be respectively given by $f(x)=|x|+1$ and $g(x)=x^2+1$. Define $h:R\to R$ by $h(x)=\begin{cases}\max\{f(x),g(x)\} & \text{if } x\le 0\\ \min\{f(x),g(x)\} & \text{if } x>0\end{cases}$ Then number of points at which $h(x)$ is not differentiable is

KEY SHEET (LECTURE SHEET)

EXERCISE - I

LEVEL - I

1) 3	2) 1	3) 1	4) 1	5) 2	6) 4	7) 2	8) 4
9) 4	10) 4	11) 4	12) 4	13) 1	14) 2	15) 2	16) 2
17) 3	18) 3	19) 1	20) 2	21) 1	22) 3	23) 3	24) 3
25) 3	26) 2	27) 2					

LEVEL - II

1) b	2) d	3) b	4) b	5) d	6) a	7) a	8) ab
9) abd	10) abc	11) bc	12) ad	13) bcd	14) ac	15) acd	16) bd
17) cd	18) ac	19) abcd	20) ad	21) ac	22) ac	23) bc	24) abcd
25) bd	26) b	27) a	28) b				

EXERCISE - II

LEVEL - I

1) 3	2) 1	3) 3	4) 3	5) 2	6) 3	7) 2	8) 3
9) 1	10) 3	11) 1	12) 4	13) 2	14) 4	15) 4	16) 1
17) 2	18) 2	19) 1	20) 2	21) 3			

LEVEL - II

1) a	2) d	3) a	4) d	5) a	6) d	7) d	8) d
9) d	10) b	11) a	12) bcd	13) ab	14) bd	15) d	16) ab
17) ad	18) cd	19) ac	20) ac	21) bcd	22) c	23) b	24) c
25) c	26) d	27) d	28) 3				